

Bellabeat Case Study

An app that focusses on women health

Agenda

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Introduction to Bellabeat



Overview

Bellabeat is a high-tech manufacturer of health-focused products for women. A small but successful company with the potential to become a significant player in the global smart device market.

Bellabeat Product Line

Bellabeat App: Health data related to activity, sleep, stress, menstrual cycle, and mindfulness habits.

Leaf: Classic wellness tracker worn as a bracelet, necklace, or clip.

Time: Wellness watch with a classic timepiece look, tracking activity, sleep, and stress.

Spring: Smart water bottle tracking daily water intake.

Bellabeat Membership: Subscription-based program offering personalized guidance on nutrition, activity, sleep, health and beauty, and mindfulness.

Problem Statement



Challenges

- Bellabeat seeks to understand current trends in smart device usage to improve their products and marketing strategies.
- The need to gain insights into how people are using non-Bellabeat smart devices.
- Desire to leverage this information to become a larger player in the global smart device market.

Objectives

- Analyze usage data from Fitbit and other non-Bellabeat devices.
- Extract meaningful insights to inform Bellabeat's marketing strategy.

Business Questions

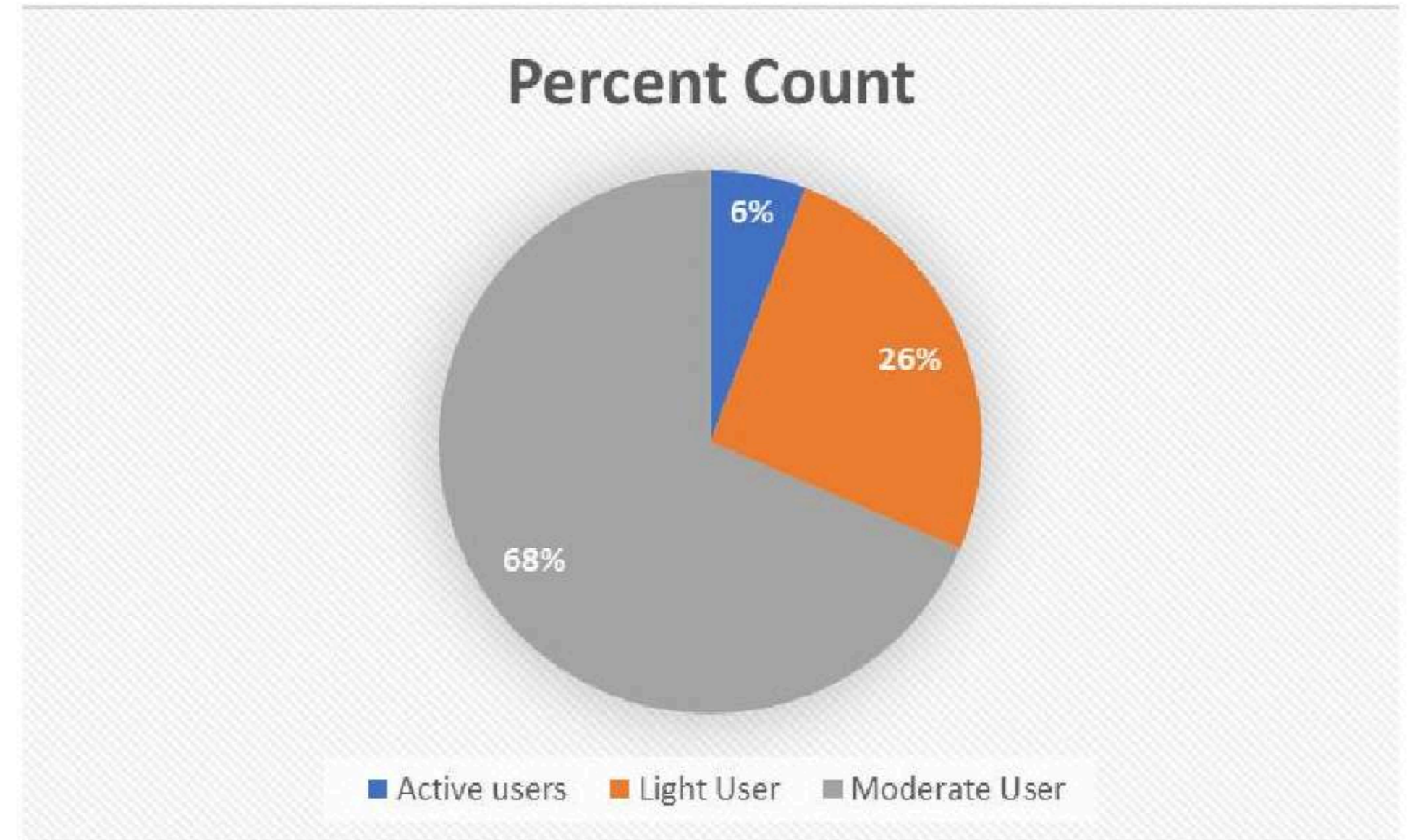
- What are some trends in smart device usage?
- How could these trends apply to Bellabeat customers?
- How could these trends help influence Bellabeat's marketing strategy?

Data Insights



User Activity Frequency

- Light Users: 26% of the users fall into this category, indicating infrequent use of the Fitbit device.
- Moderate Users: A significant majority, 68.5% of users, are classified as moderate users, suggesting consistent but not intensive engagement with the device.
- Active Users: Only 5.7% of users are highly engaged, logging their activities most frequently.

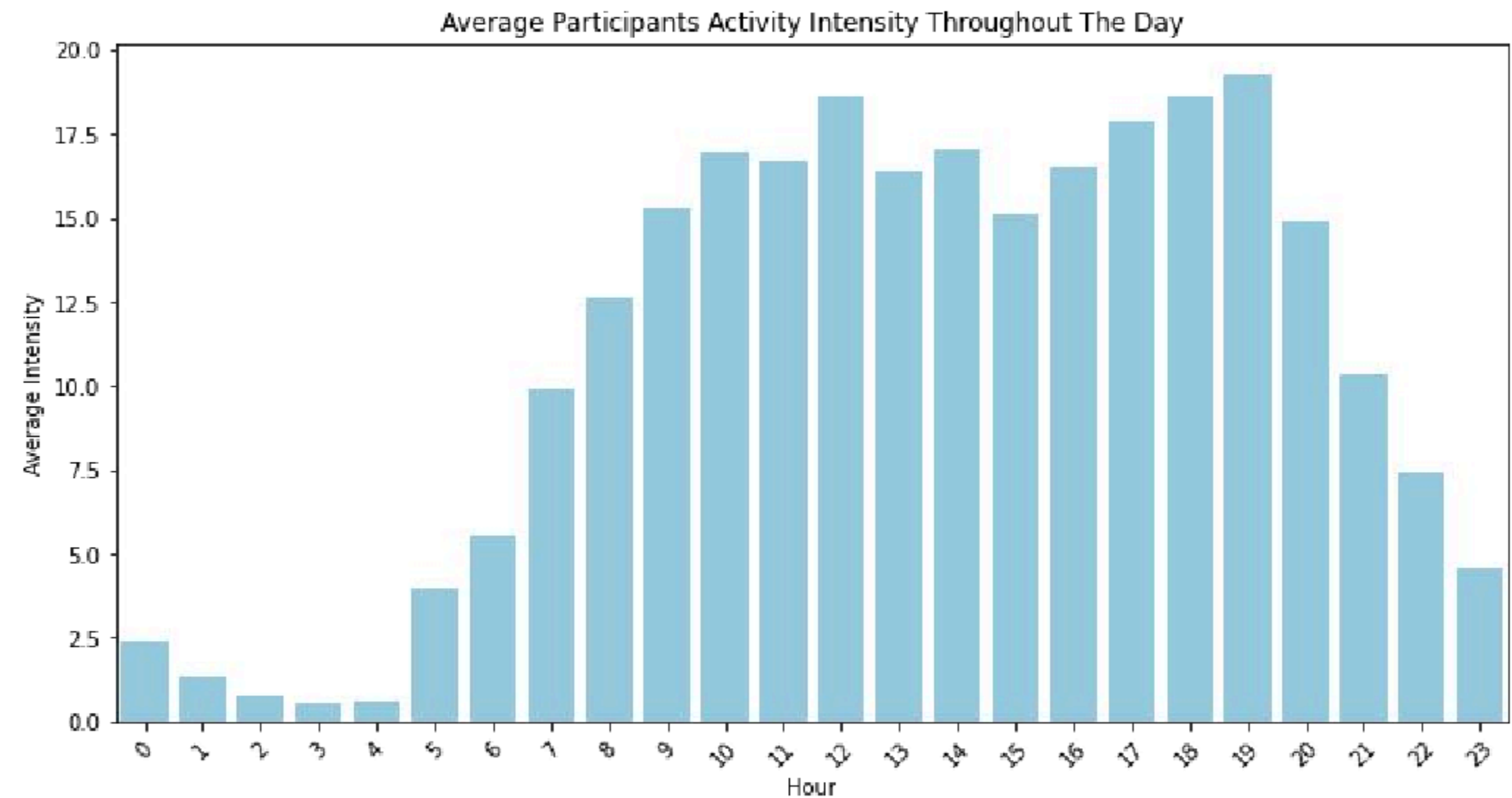


Hourly Activity Intensity

The analysis of participants' activity intensity throughout the day reveals distinct patterns:

- Peak Activity Periods:
 - Late afternoon to early evening (17:00 to 19:00)
 - Midday (12:00 to 14:00)
- Least Active Periods:
 - Late night (01:00 to 04:00)
 - Early morning and late-night hours (as expected).

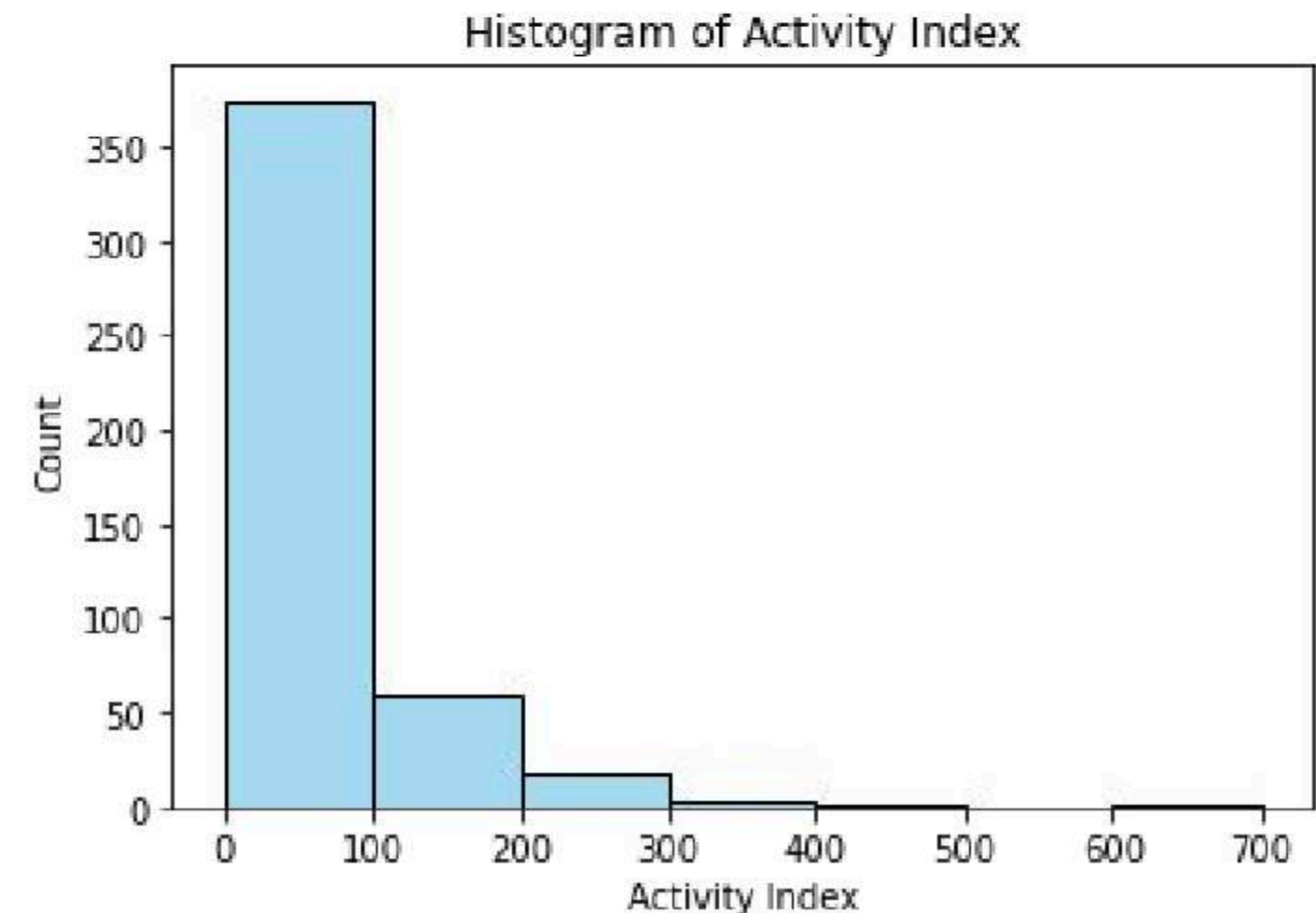
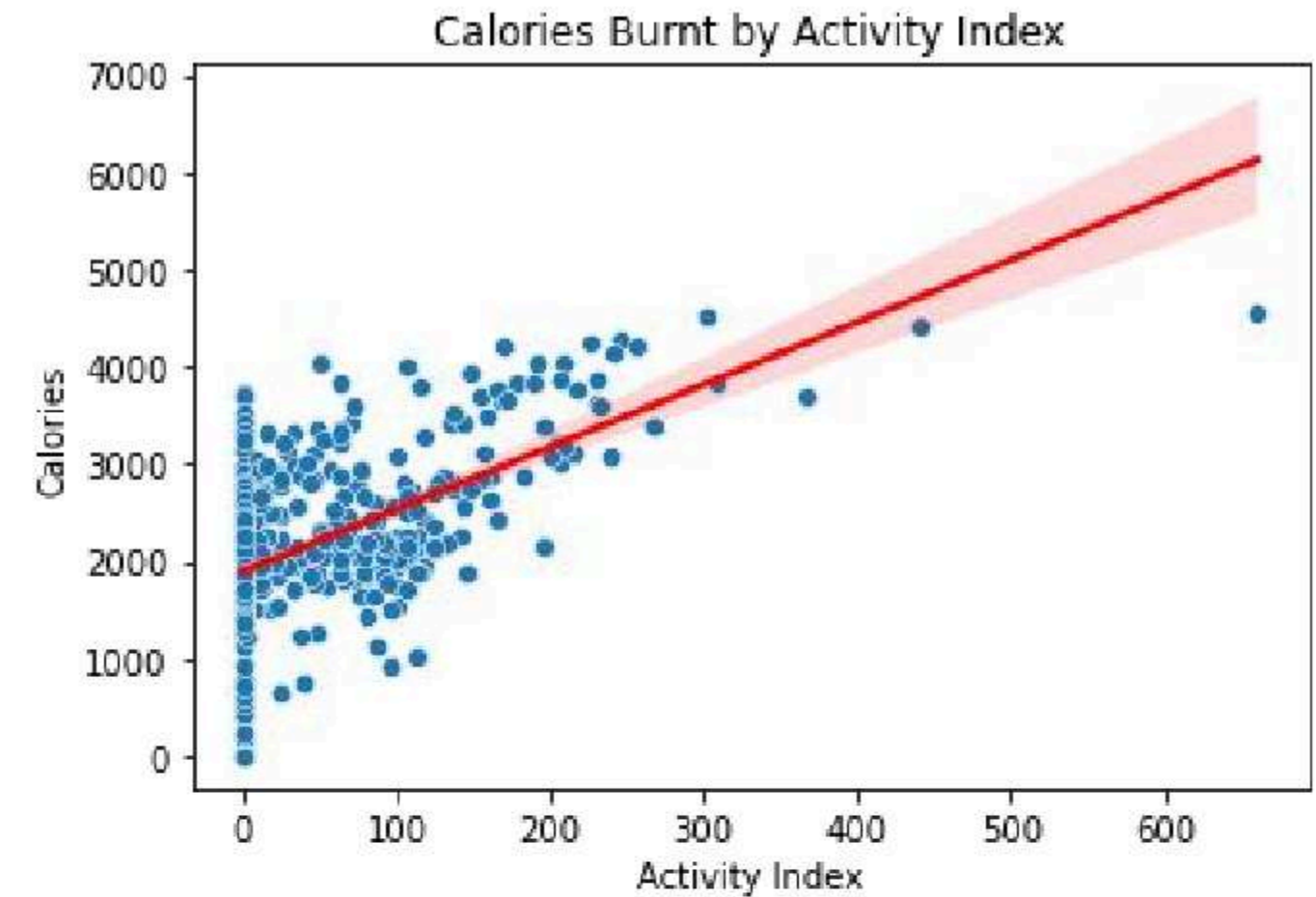
These insights can be used to tailor fitness recommendations and interventions to encourage more balanced activity throughout the day, potentially improving overall health outcomes.



Calorie vs Activity Index

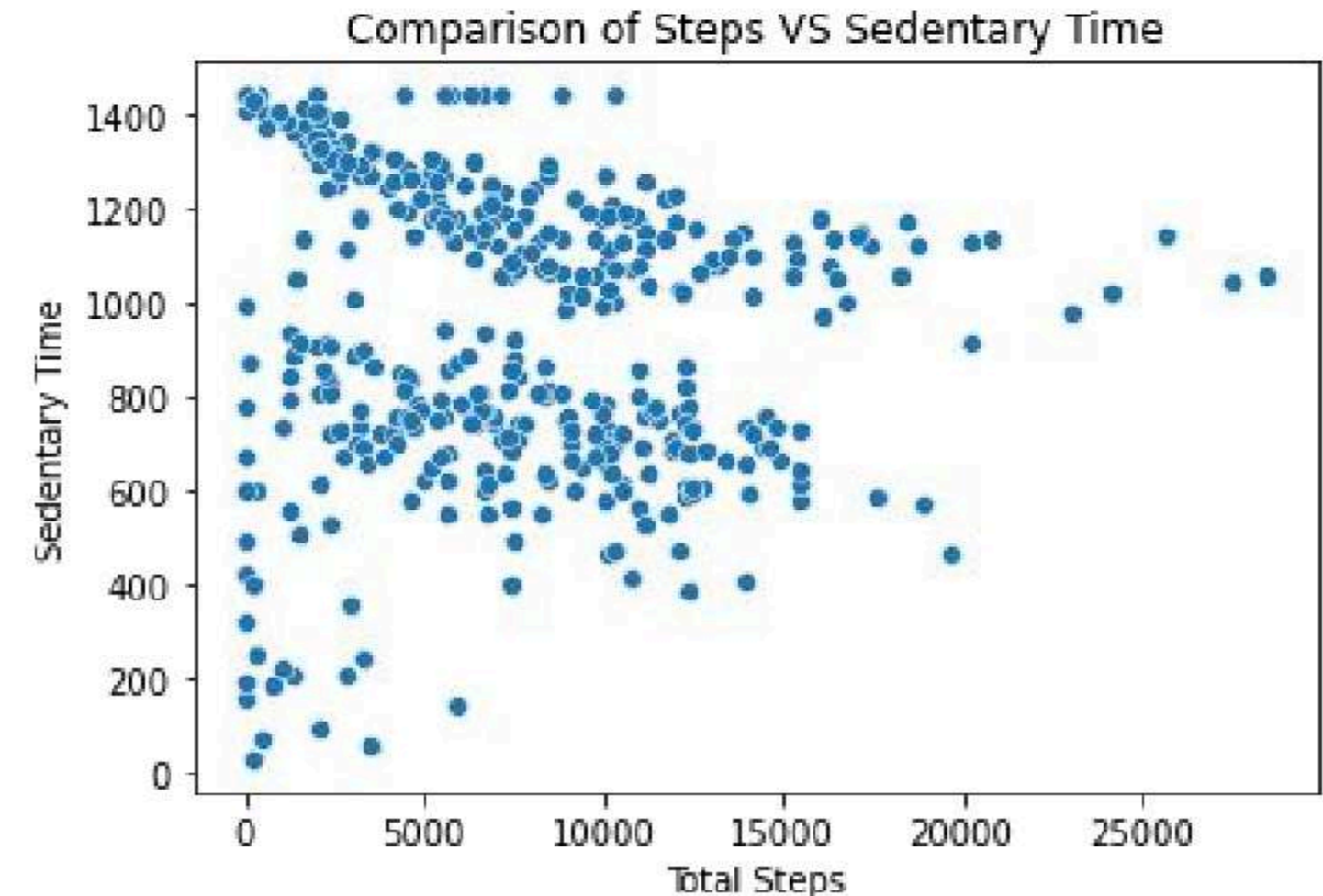
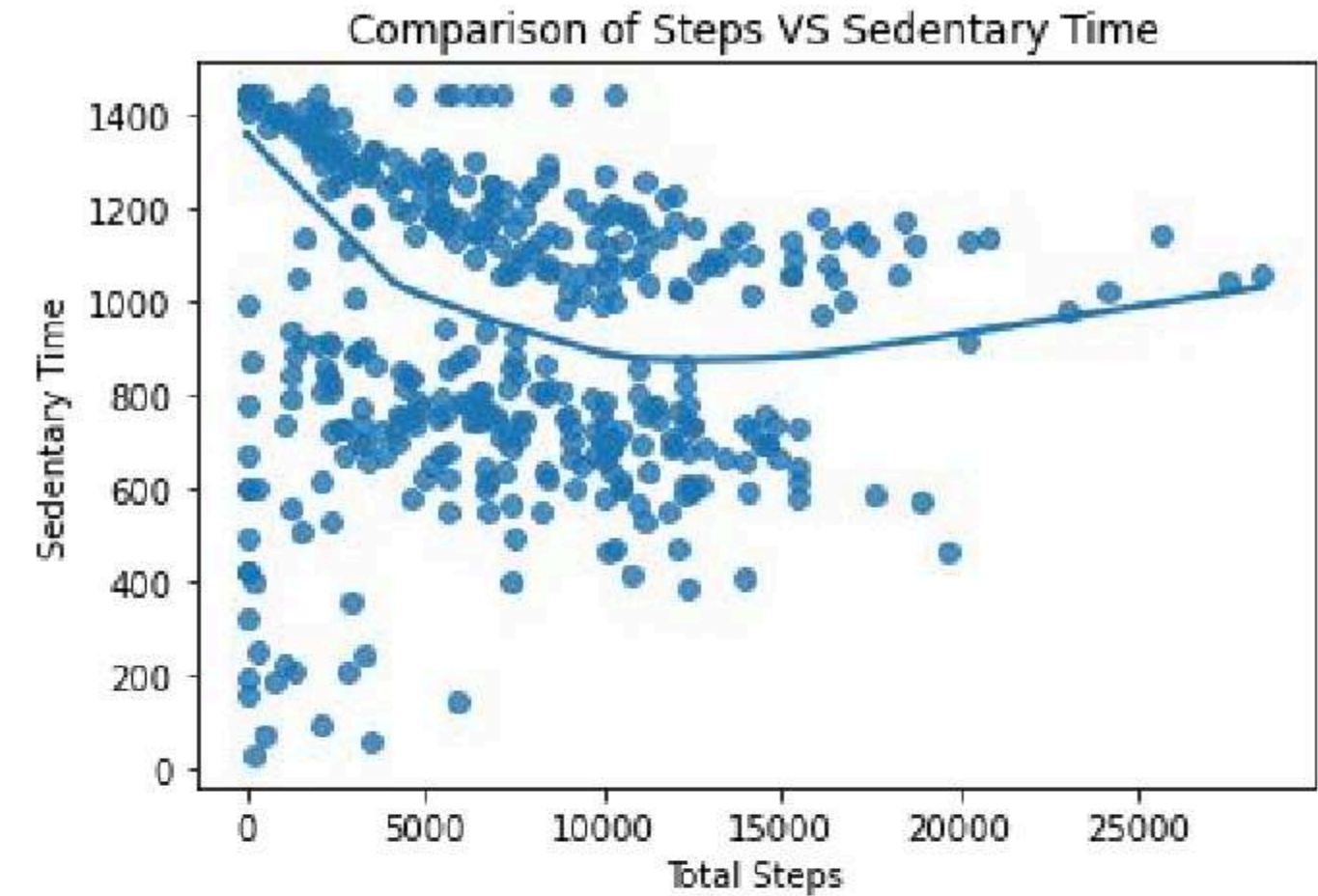
- Positive Correlation: There is a positive correlation between calorie expenditure and the intensity of physical activity, confirming that more intense activities lead to higher calorie burn.
- Low to Moderate Activity Levels: The majority of participants show low to moderate activity levels, indicating a significant opportunity to increase calorie burning by engaging in more vigorous physical activities.
- Room for Improvement: To optimize calorie expenditure, participants should be encouraged to increase their activity intensity.

Activity Index = (Very Active Minutes *2) + Fairly Active Minutes



Steps vs Sedentary Time

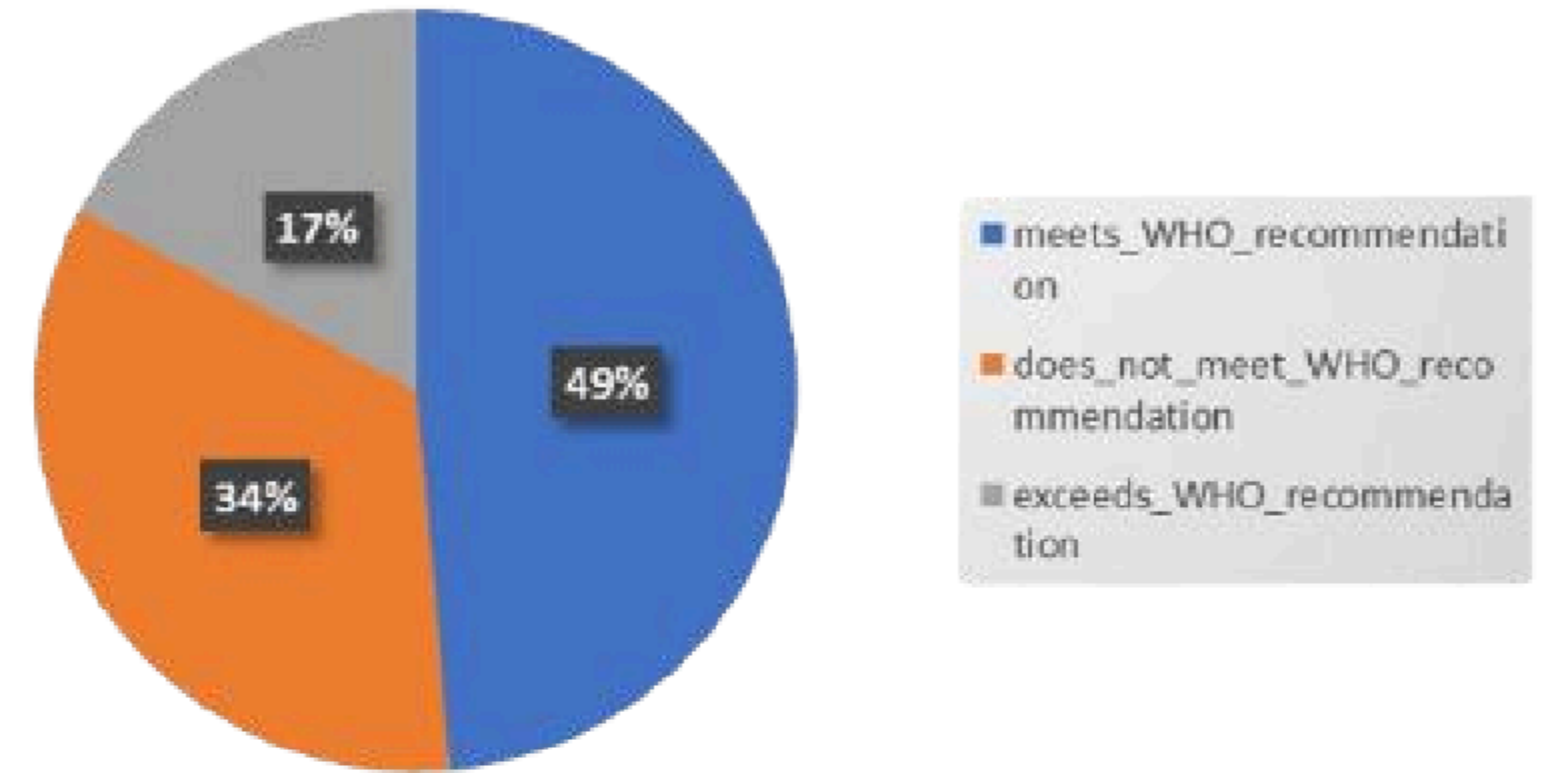
- Step Count Distribution: The majority (75%) of participants recorded a step count below 10,198.
- Sedentary Behavior Variability: There is significant variability in sedentary behavior among individuals.
 - Median Sedentary Time: About half of the participants reported sedentary times exceeding 1,057 minutes per day, equivalent to more than 16 hours.
 - High Sedentary Timex: A considerable portion of individuals reported sedentary times of up to 1,440 minutes, corresponding to more than 23 hours per day.
 - This indicates that many participants spend almost their entire day in low-intensity or sedentary activities.



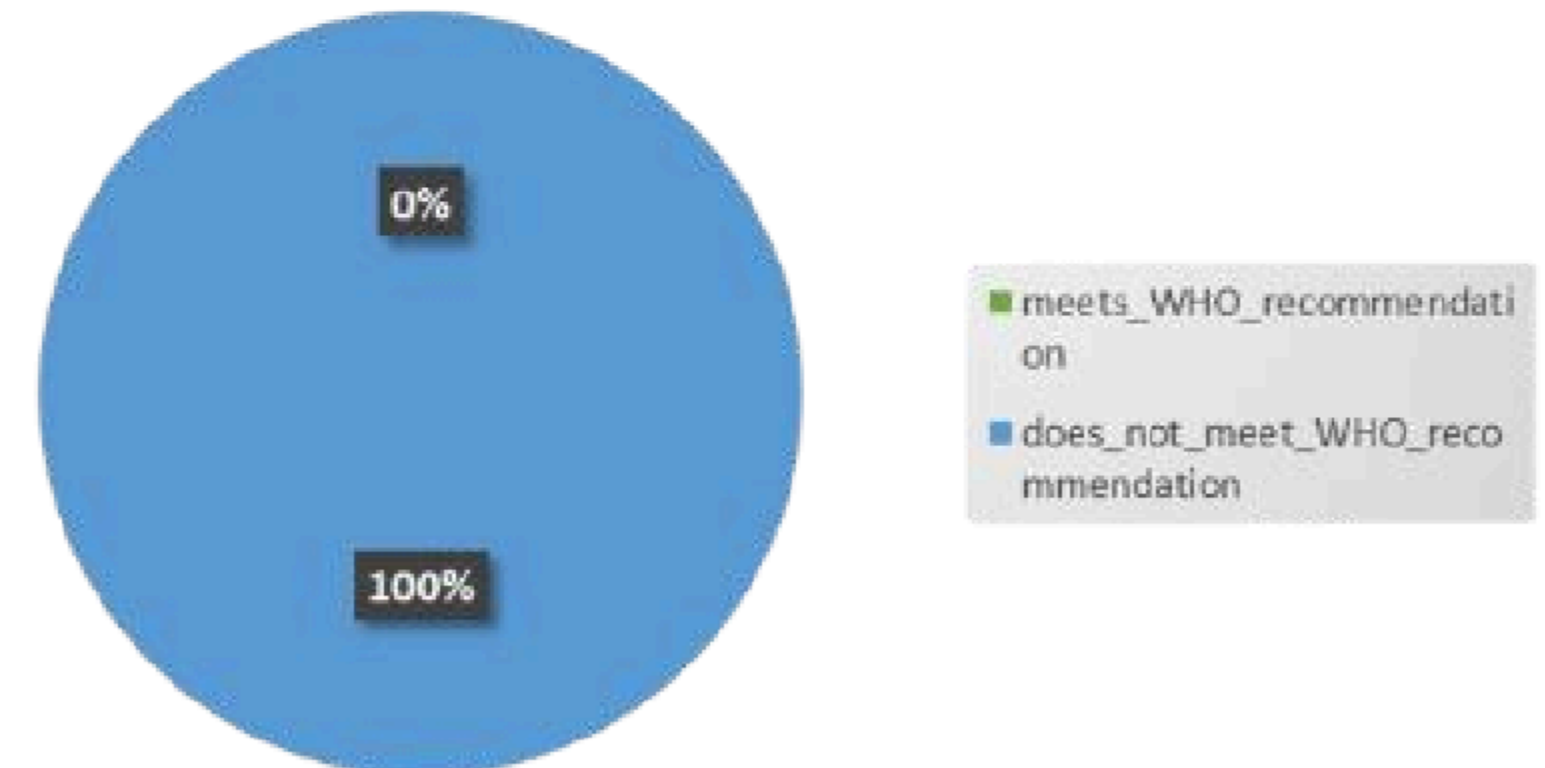
WHO Physical Activity Recommendations Analysis

The World Health Organization (WHO) recommends that adults aged 18–64 engage in at least 150–300 minutes of moderate-intensity aerobic physical activity weekly. This analysis evaluates whether users meet this recommendation.

- **Activity Levels:** When considering all activity levels (very active, fairly active, and lightly active), 17 users meet the WHO recommendations, and 6 exceed them.
- **Intensity Matters:** Excluding lightly active minutes reveals that none of the users meet the minimum WHO recommendation of 150 minutes and engage in enough moderate to vigorous physical activity.



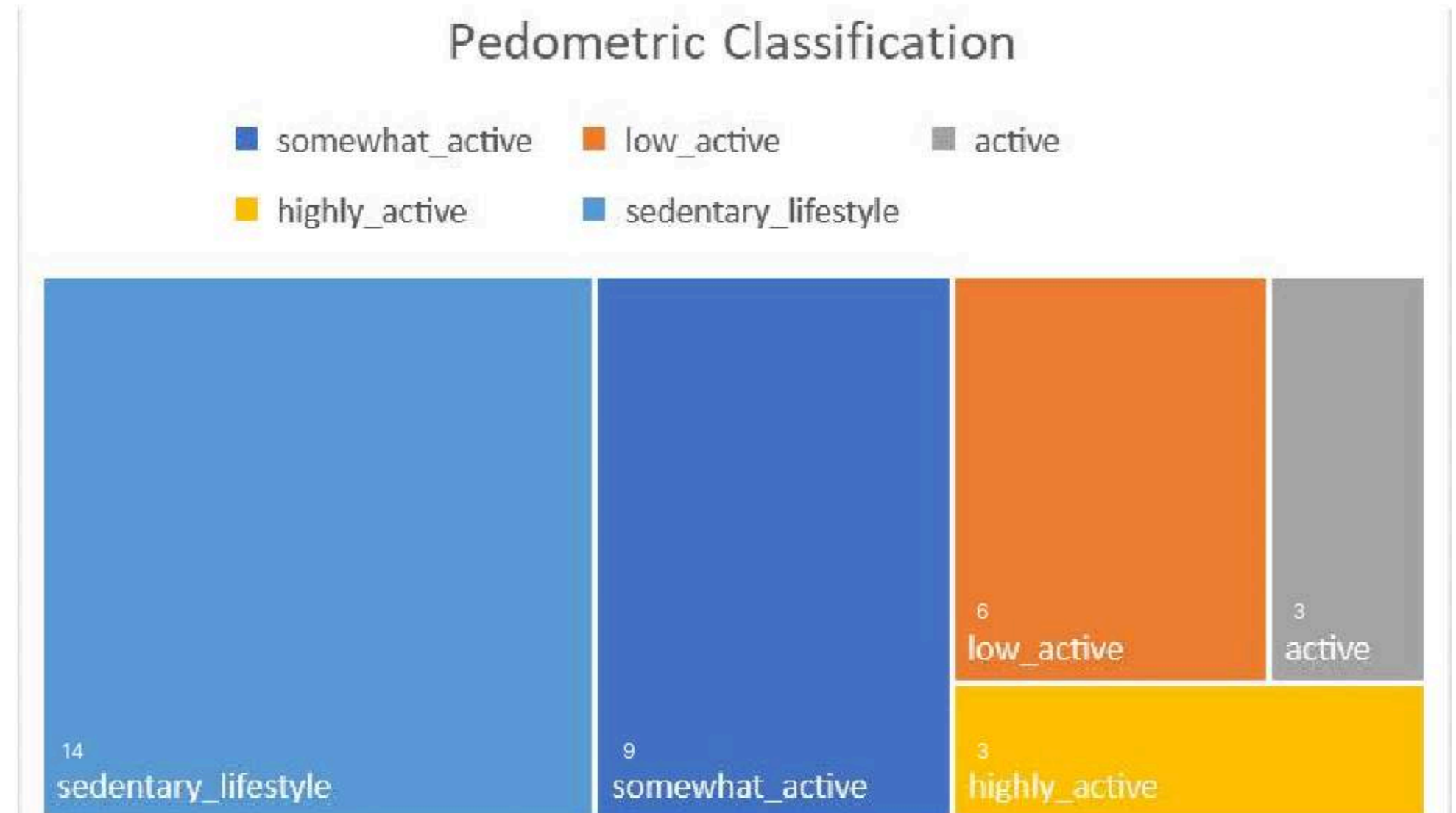
Pie Chart when considering all activity levels (very active, fairly active, and lightly active)



Pie Chart after excluding lightly active minutes

Pedometric Classification

- Sedentary Minutes: On average, users spend the most time in sedentary activities, with an average of 1384 minutes. This suggests a predominantly sedentary lifestyle among users and therefore need to encourage more active lifestyles could be beneficial.
- Lightly Active Minutes: Users spend an average of 163.33 minutes in lightly active activities, indicating some level of movement but not intensive physical activity.
- Fairly Active Minutes: The highest average for fairly active minutes is 1.67 minutes, showing minimal engagement in moderate physical activities.
- Very Active Minutes: The highest average for very active minutes is just 0.5 minutes, indicating very limited engagement in vigorous physical activities.



Conclusion



Majority of users have a sedentary lifestyle, averaging below 5,000 steps per day, with most falling within the low to somewhat active range (5,000 to 9,999 steps per day). A small percentage of users are highly active, exceeding 12,500 steps per day.

17 users meet the WHO recommendation of 150–300 minutes of moderate-intensity activity per week. However, none meet the minimum 150 minutes of moderate to vigorous activity when lightly active minutes are excluded.

Higher activity intensity correlates with higher calorie expenditure, but most participants exhibit low to moderate activity levels, indicating potential for improvement.

75% of participants record less than 10,198 steps per day, with half reporting more than 16 hours of sedentary activity per day.

Peak activity levels occur between 17:00 to 19:00 and 12:00 to 14:00, while the least active periods are during late-night hours (01:00 to 04:00).

Recommendations



Promote Active Lifestyles

- **Campaigns on Sedentary Behavior:** Launch marketing campaigns that raise awareness about the risks of a sedentary lifestyle and the benefits of regular activity. Highlight Bellabeat's features that encourage movement and track activity levels.
- **In-App Incentives:** Implement features like badges and rewards for achieving activity milestones, such as daily step goals or consistent weekly activity. These incentives can motivate users to be more active.
- **Community Events:** Sponsor or organize walking/running events and challenges to foster a sense of community among users and promote brand loyalty. Encourage users to share their progress and achievements on social media.



Targeted Notifications

- **Exercise Reminders:** Use the Bellabeat app to send personalized notifications reminding users to exercise and the benefits of high-intensity workouts, especially during their usual peak active times (12:00-14:00 and 17:00-19:00).
- **Movement Prompts:** Market the "get up and move" reminder feature to help users reduce sedentary time and complete their daily activity goals. Emphasize how this feature can help them stay active throughout the day.



Product Differentiation

- **Women-Centric Features:** Emphasize Bellabeat's focus on women's health. Introduce features such as period tracking and notifications on how to stay active during menstruation. This can help women maintain their activity levels and reach their fitness goals every day.
- **Health Professional Collaborations:** Collaborate with health professionals to promote the importance of reducing sedentary time and using Bellabeat products to achieve a healthier lifestyle.



Social Media Engagement

- **Influencer Partnerships:** Utilize social media platforms and influencer partnerships to reach target audiences interested in improving their activity levels and overall health. Encourage influencers to share their experiences with Bellabeat products and how they help in maintaining an active lifestyle.
- **Educational Content:** Create and share educational content on the importance of regular exercise, reducing sedentary time, and improving sleep health. Use blog posts, videos, and social media content to engage and inform users.



Thank You

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